

Airbnb Market Intelligence Platform

Data Engineering, Analytics & Machine Learning on the Inside Airbnb Dataset

Submitted By:

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Prepared For:

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Dataset:

Inside Airbnb – London

Technologies:

Python | DuckDB | SQL | Streamlit | ML

1. Executive Summary

This project developed an end-to-end Airbnb Market Intelligence Platform using the Inside Airbnb London dataset. The objective was to transform raw Airbnb data into a reliable analytical environment that supports business intelligence, statistical analysis, and predictive modeling.

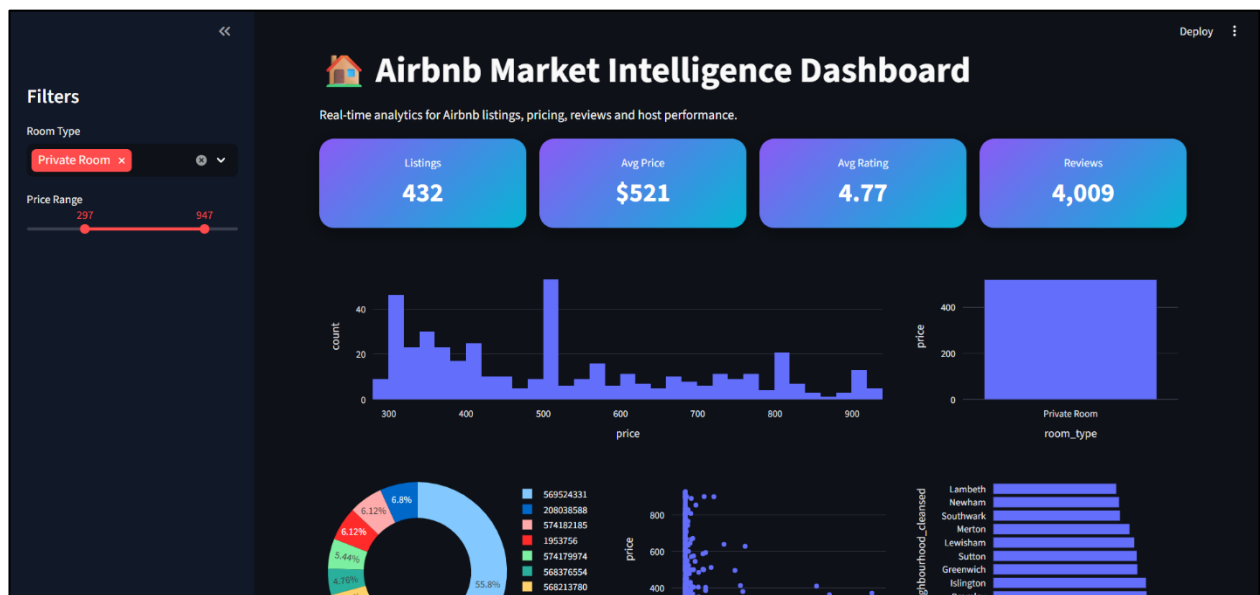
A configurable data engineering pipeline was built to automate data ingestion, cleaning, validation, enrichment, and warehouse loading. The solution includes logging, retry mechanisms, incremental processing, metadata tracking, and data lineage documentation to improve reliability and maintainability.

The processed datasets were integrated into a unified listing master table and modeled using a dimensional star schema in DuckDB. Key business metrics such as occupancy rate, estimated revenue, host tenure, review frequency, and price-per-bedroom were derived to support analytical use cases.

Exploratory Data Analysis identified significant pricing differences across room types and neighbourhoods, while host analysis revealed a concentration of listings among a relatively small number of hosts. Statistical testing and regression analysis were conducted to validate key marketplace hypotheses and identify the main drivers of listing prices.

Machine learning models were developed for price prediction and evaluated using standard performance metrics. An interactive Streamlit dashboard was also created to provide stakeholders with real-time access to market insights, pricing trends, host performance indicators, and geographic patterns.

The project demonstrates how data engineering, analytics, and machine learning can be combined to generate actionable insights that support pricing optimization, market monitoring, and strategic decision-making in the Airbnb marketplace.



[Figure 1 : Dashboard Overview]

2. Objectives & Scope

➤ Project Objectives

This project aimed to transform raw Airbnb data into a reliable analytical platform capable of supporting business intelligence, statistical analysis, and machine learning use cases.

The primary objectives were:

- Build a repeatable data engineering pipeline
- Create analytics-ready datasets
- Design a dimensional warehouse model
- Perform exploratory and statistical analysis
- Develop predictive machine learning models
- Deliver an interactive business dashboard

➤ Scope

The analysis focused on the London Airbnb dataset obtained from Inside Airbnb.

Prioritization Strategy

Priority was given to:

1. Data Engineering
2. Exploratory Analysis
3. Statistical Analysis
4. Machine Learning
5. Interactive Dashboard

3. Dataset Overview

➤ 3.1 Dataset Source

<https://insideairbnb.com/get-the-data/> — City-level data files

➤ 3.2 Files Used

File	Purpose
listings.csv	Listing metadata
calendar.csv	Daily availability & pricing
reviews.csv	Guest reviews
neighbourhoods.csv	Geographic classification

➤ 3.3 Dataset Size Summary

Dataset	Rows	Columns
listings	96871	79
calendar	35357974	7
reviews	2097996	6
neighbourhoods	33	2

➤ 3.4 Schema Overview

Column	Data Type	Sample Value
id	int64	12345
host_id	int64	56789
price	object	\$120
room_type	object	Entire home

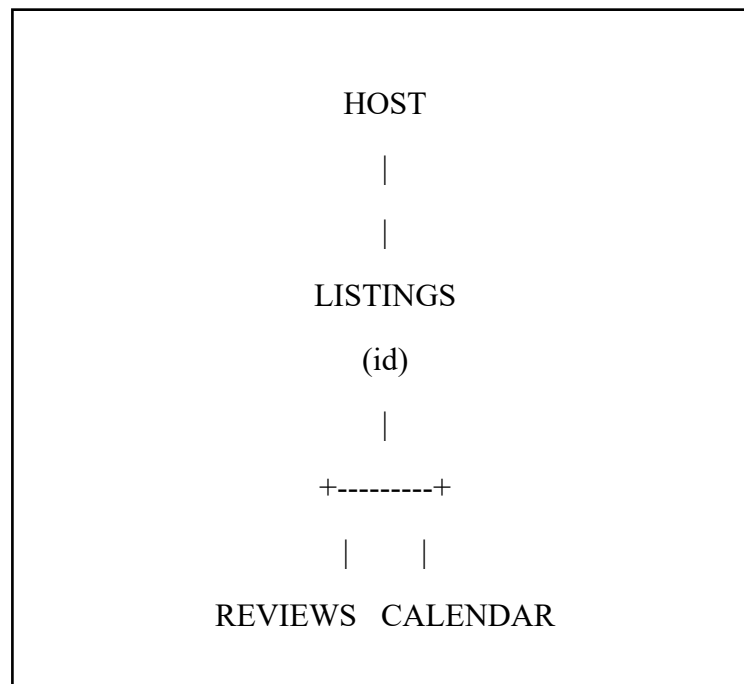
➤ 3.5 Dataset Relationships

Primary Keys:

- listings: id - unique identifier for each property
- calendar: (listing_id, date) - composite key, one row per listing per date
- reviews: id - unique identifier for each review

Foreign Keys:

- calendar.listing_id -> listings.id
- reviews.listing_id -> listings.id



[Figure 2: Entity Relationship Diagram]

➤ 3.6 Missing Value Assessment

- **Listings**

```
license          100.000000
calendar_updated 100.000000
neighbourhood_group_cleansed 100.000000
neighbourhood_overview 57.460953
neighbourhood     57.459921
host_neighbourhood 52.669013
host_about        48.557360
beds              36.047940
price             36.035552
estimated_revenue_l365d 36.035552
bathrooms         35.971550
host_response_rate 32.731158
host_response_time 32.731158
host_acceptance_rate 28.656667
review_scores_location 24.946578
review_scores_value 24.946578
review_scores_checkin 24.945546
review_scores_communication 24.921803
```

[Figure 3 : Missing Values in listings]

- **Calendar**

```
price          100.0
adjusted_price 100.0
listing_id      0.0
available       0.0
date            0.0
minimum_nights  0.0
maximum_nights  0.0
dtype: float64
```

[Figure 4 : Missing Values in calendar]

- **Reviews**

```

comments      0.009581
reviewer_name  0.000095
id             0.000000
listing_id    0.000000
reviewer_id    0.000000
date          0.000000
dtype: float64

```

[Figure 5 : Missing Values in reviews]

- **Neighbourhood**

```

neighbourhood_group  100.0
neighbourhood         0.0
dtype: float64

```

[Figure 6 : Missing Values in neighbourhood]

➤ 3.7 Data Dictionary Review

Column	Meaning
minimum_nights	Minimum stay required
availability_365	Available days per year
host_since	Host registration date

➤ 3.8 Dataset Limitations

- Dataset is scraped from Airbnb rather than provided directly by Airbnb.
- Availability may not always indicate bookings.
- Dataset represents a snapshot in time.
- Some listings contain incomplete information.
- Historical booking transactions are unavailable.

➤ 3.9 Assumptions

- Prices represent nightly rates.
- Review count can be used as a proxy for demand.
- Unavailable dates may indicate booked or blocked listings.
- Hosts with multiple listings are treated as commercial operators.

➤ 3.10 Business Context

- Listing

A property advertised on Airbnb.

- Host

Owner or manager responsible for one or more listings.

- Review

Customer feedback after a stay.

- Calendar

Daily pricing and availability information.

- Neighbourhood

Geographical grouping used for market analysis.

4. Methodology

The project followed a structured data lifecycle consisting of:

1. Data Ingestion
2. Data Profiling
3. Data Cleaning
4. Data Enrichment
5. Data Modeling
6. Exploratory Analysis
7. Statistical Analysis
8. Machine Learning
9. Dashboard Development



[Figure 7 : End-to-End Workflow]

5. Engineering Approach

➤ 5.1 Data Ingestion & Profiling

A repeatable ingestion pipeline was developed using Python and Pandas. The pipeline loads Airbnb datasets from the raw data layer and performs automated profiling.

Profiling included:

- Row counts
- Column counts
- Missing value assessment
- Duplicate detection
- Data type analysis
- Cardinality assessment

➤ 5.2 Data Cleaning & Standardization

Duplicate Analysis

Duplicate records were assessed using deterministic matching based on complete row equality.

No significant duplicate records were observed within the primary datasets.

Future enhancements may include fuzzy matching techniques to identify semantically similar listings based on title and description similarity.

Outlier Analysis

Outlier detection was performed using the Interquartile Range (IQR) method.

Several listings exhibited exceptionally high nightly prices compared to the market median. These records likely represent luxury or premium properties rather than data quality issues.

Availability and review count distributions also contained high-value outliers, which were retained because they represent legitimate business scenarios.

Data Validation

Validation rules were applied to ensure compliance with domain expectations.

The following checks were performed:

- Price must be non-negative.
- Latitude values must fall between -90 and 90.
- Longitude values must fall between -180 and 180.
- Review counts must be non-negative.

No critical validation failures were identified.

Data Cleaning

The following cleaning steps were performed:

- Removed currency symbols from price fields.
- Converted prices into numeric format.
- Standardized date fields into datetime format.
- Standardized room type labels.
- Removed columns containing 100% missing values.
- Imputed review score attributes using median values.

These transformations improved consistency and analytical readiness.

➤ 5.3 Data Enrichment & Joining

- **Occupancy Rate Calculation**

To estimate listing demand, calendar availability data was transformed into an occupancy metric. The Airbnb calendar dataset records whether a listing is available on a given date. Listings marked as unavailable (available = 'f') were assumed to be booked.

A binary feature named booked was created:

Available Value	Booked Value
t	0
f	1

- **Engineering Decision**

The assignment dataset does not provide direct booking information. Therefore, unavailable calendar dates were treated as booked nights. While this assumption may slightly overestimate occupancy because hosts can manually block dates, it is widely used in Airbnb market analysis and provides a reasonable estimate of demand.

Output

A new dataset (occupancy.csv) was generated containing:

- listing_id
- occupancy_rate

This dataset will be joined with listing-level information during later enrichment stages.

- **Neighbourhood-Level Aggregation**

Neighbourhood statistics were calculated and joined to the master dataset. Metrics included median listing price, listing density, and average review rating, providing additional geographic and market-level context.

- **Host Tenure**

Host tenure was calculated as the number of years between the host's registration date and the current date. This feature provides insight into host experience and platform engagement.

- **Review Frequency**

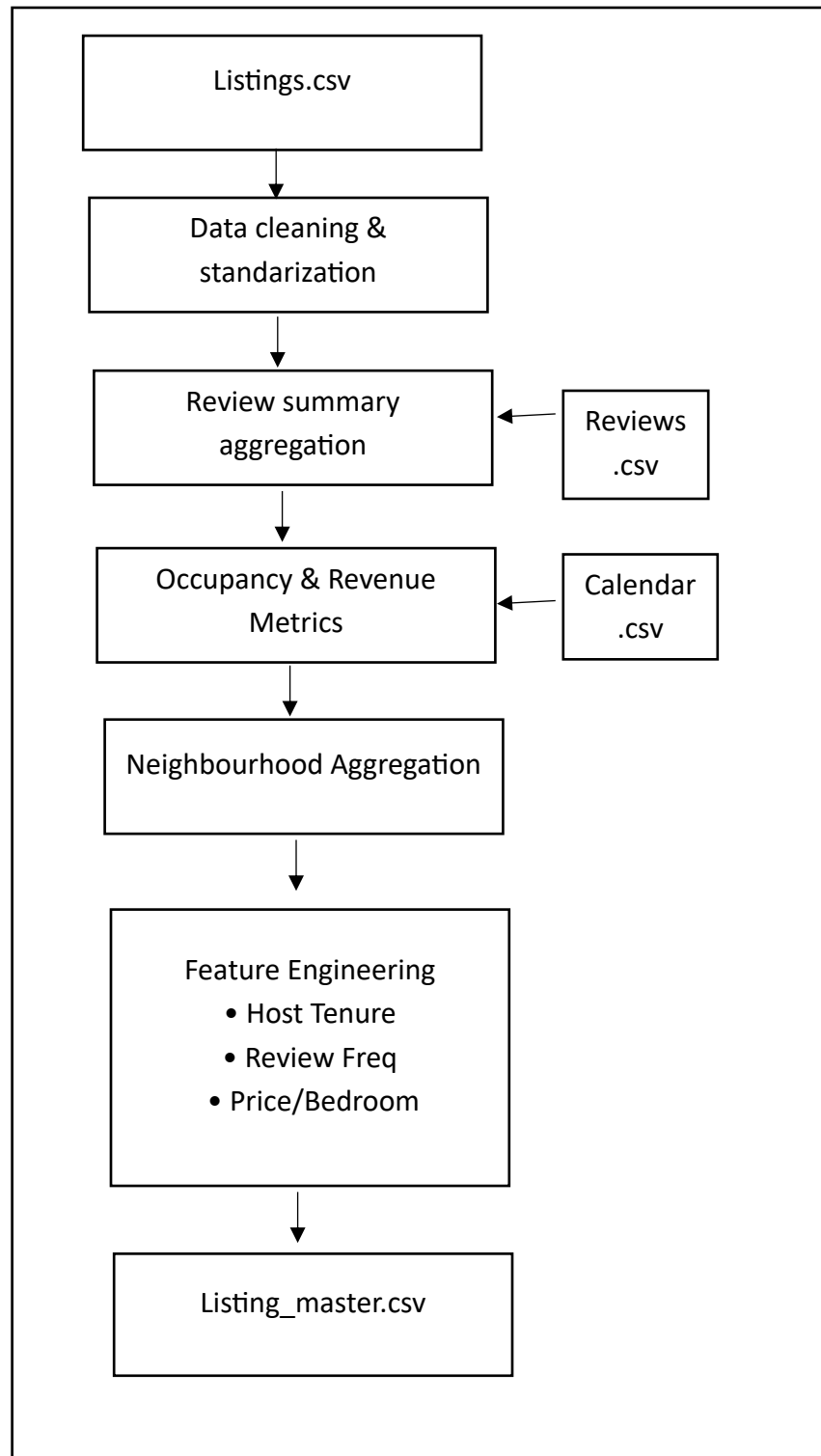
Review frequency was calculated as the number of reviews received per year of host activity. This metric acts as a proxy for listing demand and guest engagement.

- **Price Per Bedroom**

Price per bedroom was derived by dividing the listing price by the number of bedrooms. This feature enables standardized comparison between properties of different sizes.

- **Enriched Listing Master Table**

Multiple datasets were integrated to create an enriched listing master table. Review summaries, occupancy metrics, revenue estimates, neighbourhood-level statistics, and derived business features were combined into a single analytics-ready dataset. This dataset serves as the foundation for subsequent exploratory analysis and business insights.



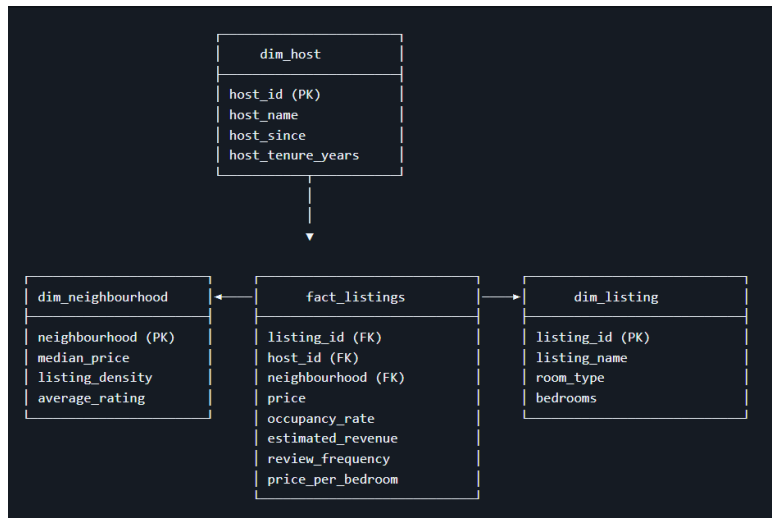
[Figure 8 : Data Enrichment Pipeline]

➤ 5.4 Data Modeling

Star Schema Design

A dimensional model was designed using a star schema architecture. Listing performance metrics were stored in a central fact table, while descriptive attributes were organized into separate dimension tables for hosts, listings, and neighbourhoods.

This design improves query performance and simplifies analytical reporting.



[Figure 9 : Star Schema Design]

Data Modeling

A dimensional star schema was implemented using DuckDB to support analytical reporting and business intelligence use cases. The model consists of one fact table (fact_listings) and three dimension tables (dim_host, dim_listing, and dim_neighbourhood).

Fact Table

- **fact_listings**

Measures stored:

- Price
- Occupancy Rate
- Estimated Revenue
- Review Frequency
- Price Per Bedroom

Dimension Tables

- **dim_host**

Host information and tenure

- **dim_listing**

Listing attributes such as room type and bedrooms

- **dim_neighbourhood**

Neighbourhood-level aggregated metrics

Relationships

Fact Table	Dimension Table	Key
fact_listings	dim_host	host_id
fact_listings	dim_listing	id
fact_listings	dim_neighbourhood	neighbourhood_cleansed

Analytical Use Cases

The model supports:

- Revenue analysis
- Occupancy analysis
- Host performance analysis
- Neighbourhood comparisons
- Room-type performance analysis

Modeling Trade - offs

A star schema was selected to simplify analytical queries and improve reporting performance. Some data redundancy was accepted to reduce join complexity. Slowly changing dimensions were not implemented because the dataset represents a static snapshot rather than a continuously changing operational system.

```
$ python src/modeling.py
neighbourhood_cleansed avg_revenue
0 Lambeth 21777.186104
1 Westminster 16808.500651
2 Kensington and Chelsea 15366.722525
3 City of London 13741.300459
4 Camden 13074.275289
5 Islington 11709.915094
6 Southwark 10618.879117
```

[Figure 10 : Modeling Trade - offs]

➤ 5.5 Pipeline Design & Automation

○ Configurable Pipeline

A configuration-driven pipeline was implemented. Dataset locations and processing parameters are managed through a central configuration file, enabling the same pipeline to process multiple cities with minimal code changes.

- **Logging**

A logging framework was integrated into the pipeline to capture execution events, processing status, and errors. Log records are written to a centralized log file for monitoring and troubleshooting.

- **Retry Logic**

A retry mechanism was added to automatically reattempt failed operations before marking them as unsuccessful.

- **Incremental Processing**

The pipeline tracks the last successful execution date and processes only newly available data.

- **Metadata Management**

Processing metadata such as dataset name, status, row counts, and timestamps are recorded for auditing and monitoring.

- **Data Lineage**

Lineage documentation was maintained to trace the flow of data from raw source files through transformations to final analytical outputs.

- **Pipeline Automation**

The pipeline is configuration-driven, reusable across cities, and incorporates logging, error handling, retry mechanisms, metadata tracking, and lineage documentation.

6. EDA Findings

➤ 6.1 Summary Statistics & Distributions

Descriptive Statistics

Descriptive statistics were generated for all major numerical attributes including price, review scores, occupancy estimates, revenue estimates, and availability measures. These statistics provide an overview of central tendency, variability, and potential outliers within the dataset.

Business Interpretation

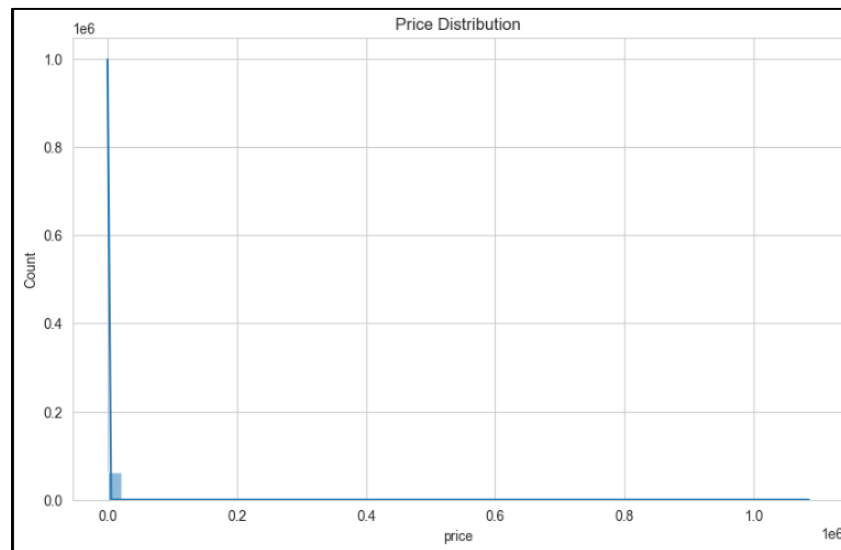
Properties exhibit substantial variation in pricing and availability, indicating a highly heterogeneous marketplace. Such variation suggests opportunities for market segmentation and differentiated pricing strategies.

1. Price Distribution

The distribution of listing prices was visualized to understand market pricing dynamics and identify potential outliers.

Business Interpretation

The price distribution is expected to be right-skewed, indicating that while most listings are relatively affordable, a smaller number of premium properties command significantly higher prices. This suggests the existence of luxury market segments alongside mainstream accommodations



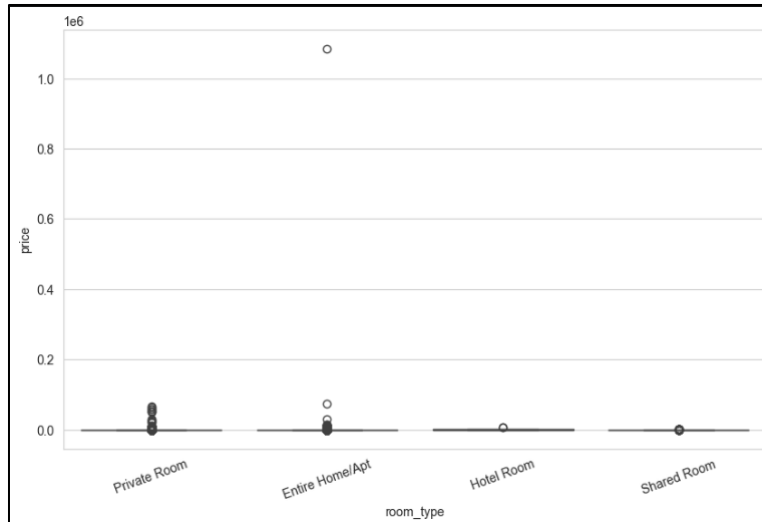
[Figure 11 : Price Distribution]

2. Price by Room Type

Pricing differences were analyzed across room types.

Business Interpretation

Entire homes generally command higher prices compared to private or shared rooms. This reflects the additional privacy and space offered by entire-property rentals and highlights the importance of room type as a pricing determinant.

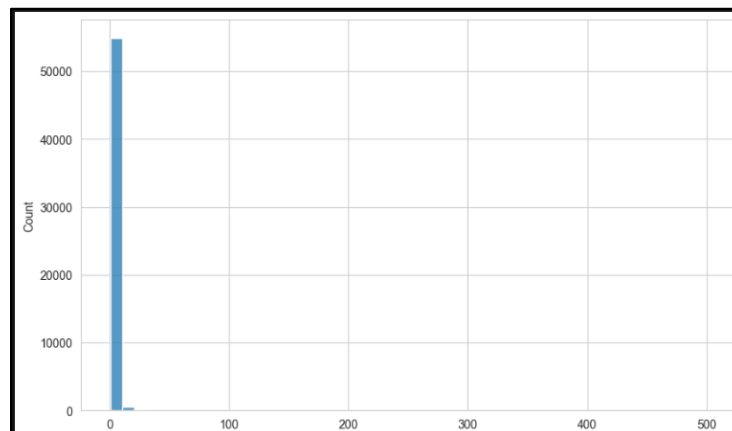


[Figure 12 : Price by Room Type]

3. Listings Per Host

Business Interpretation

Most hosts manage only a small number of listings, while a limited group controls large property portfolios. This concentration is consistent with platform marketplaces where commercial operators coexist alongside casual hosts.

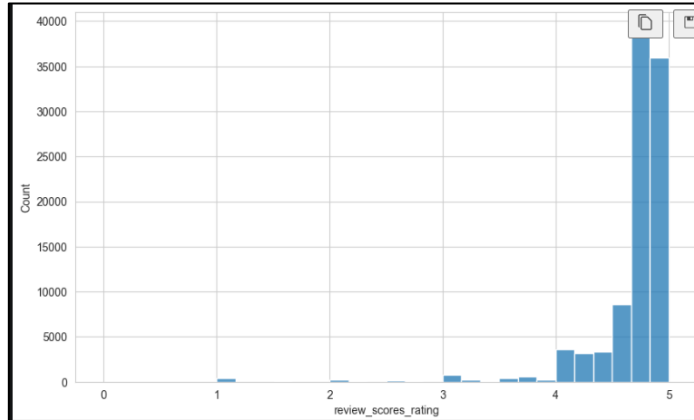


[Figure 13 : Listing per host]

4. Review Score Distribution

Business Interpretation

Review scores are typically clustered toward the upper end of the scale, suggesting potential rating inflation where most properties receive highly positive reviews.

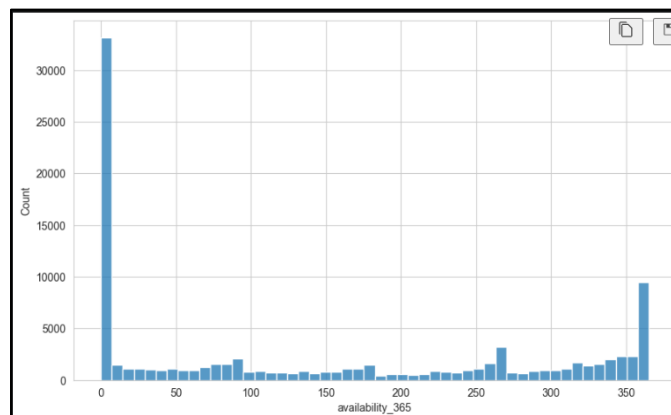


[Figure 14 : Review Score Distribution]

5. Availability Distribution

Business Interpretation

Availability patterns provide insight into occupancy levels and host strategies. Lower availability may indicate stronger demand or more aggressive booking performance.



[Figure 15 : Availability Distribution]

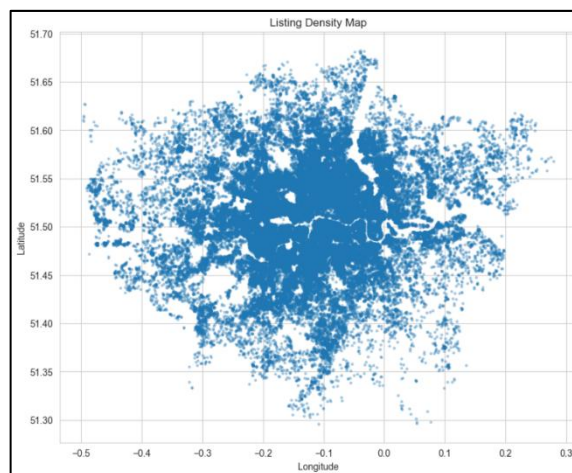
➤ 6.2 Geographic & Spatial Analysis

1. Listing Density Distribution

A spatial density visualization was created using listing coordinates to identify areas with high Airbnb concentration.

Business Interpretation

Listings are concentrated in specific geographic clusters rather than being uniformly distributed. These clusters likely correspond to popular tourist destinations, transportation hubs, and commercial districts where accommodation demand is strongest.



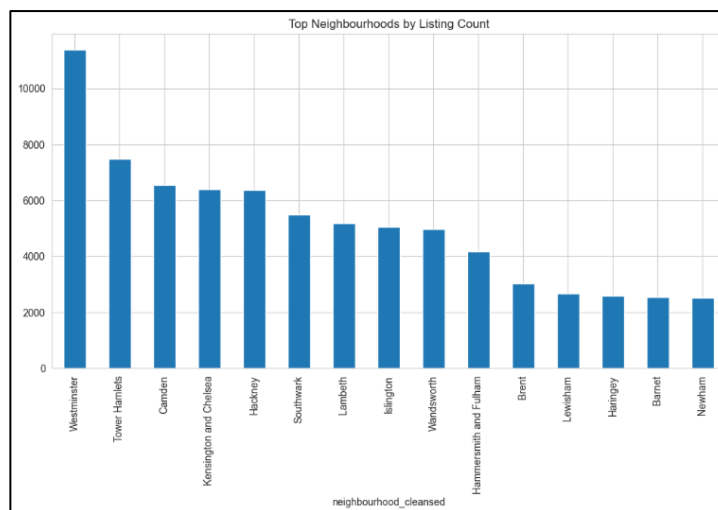
[Figure 16: Listing Density Map]

2. Neighbourhood Listing Density

Listing counts were aggregated by neighbourhood to identify areas with the highest supply concentration.

Business Interpretation

A relatively small number of neighbourhoods account for a large share of market supply. High-density areas may face stronger competition among hosts while benefiting from higher customer visibility.



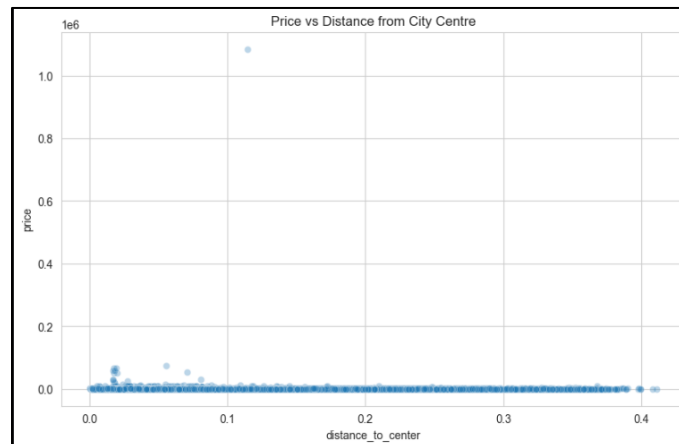
[Figure 17 : Neighbourhood Listing Density]

3. Geographic Pricing Gradient

Distances from the city center were estimated using geographic coordinates and compared with listing prices.

Business Interpretation

Properties closer to the city center generally command premium prices due to greater accessibility, tourism demand, and proximity to business districts. This relationship provides evidence of location-based pricing premiums.

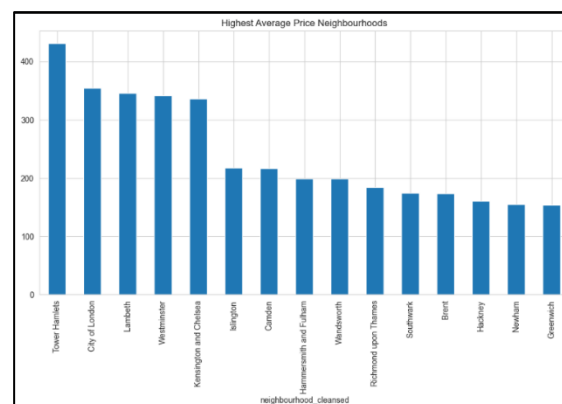


[Figure 18 : Geographic Pricing Gradient]

4. Average Price by Neighbourhood

Business Interpretation

Neighbourhoods with the highest average prices represent premium submarkets where demand exceeds supply or where guests are willing to pay more for location and amenities.



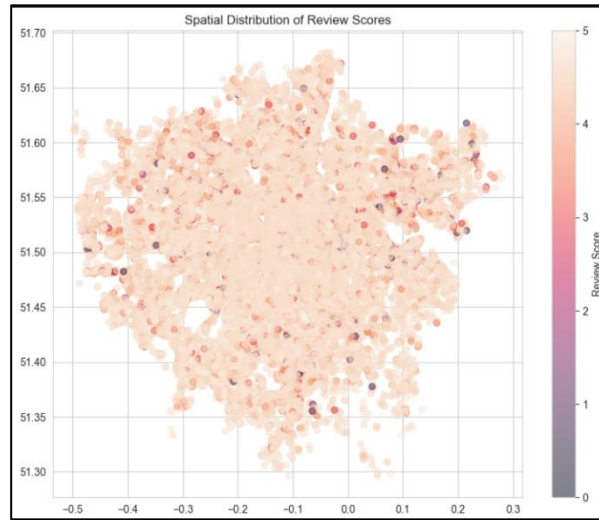
[Figure 19 : Average price by neighbourhood]

5. Spatial Review Scores

Review scores were visualized geographically to identify neighbourhood-level differences in customer satisfaction.

Business Interpretation

Clusters of consistently high ratings may indicate neighbourhoods with superior guest experiences, stronger host performance, or more desirable amenities.



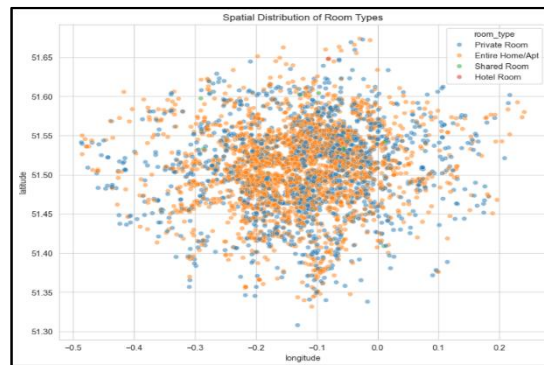
[Figure 20 : Spatial Review Scores]

6. Geographic Room Type Clustering

Room types were mapped geographically to determine whether certain accommodation types cluster in specific areas.

Business Interpretation

Entire homes may dominate residential neighbourhoods while private rooms can be more common in dense urban areas. Understanding these patterns helps explain local supply structures.

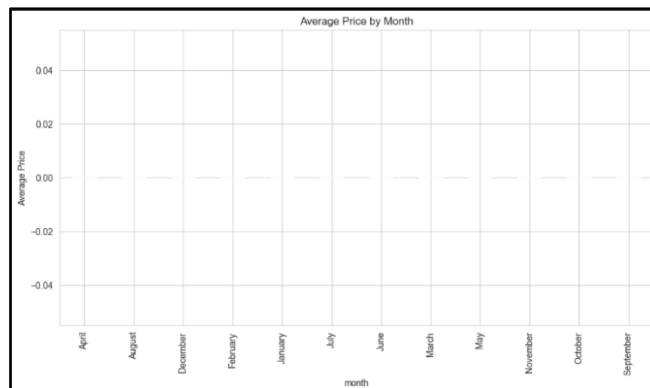


[Figure 21 : Geographical Room type by Clustering]

➤ 6.3 Temporal & Seasonal Trends

1. Seasonal Pricing Trends

Monthly average prices were analyzed to identify seasonal fluctuations and peak demand periods.



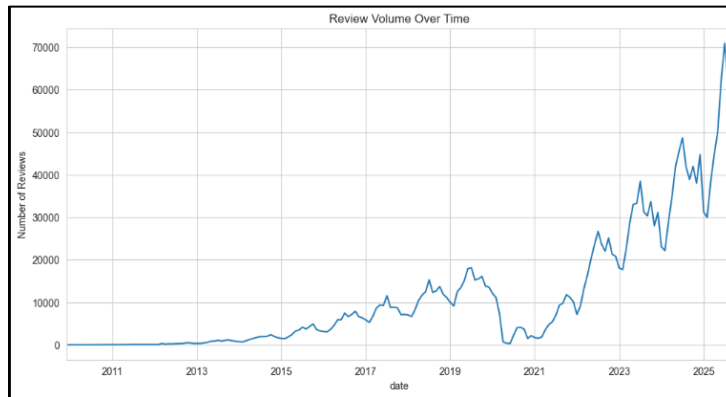
[Figure 22 : Seasonal Pricing Trends]

2. Review Volume Trends

Review activity was analyzed over time to assess growth or decline in marketplace demand.

Business Interpretation

An increasing review trend may indicate market growth and rising booking activity. Declining review volume may signal weaker demand or changing traveler behavior.



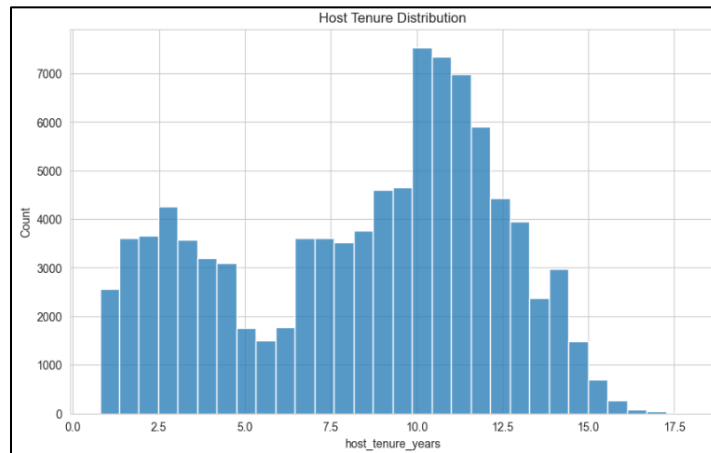
[Figure 23 : Review Volume Trends]

3. Host Tenure Distribution

Host tenure was analyzed to understand the balance between experienced and newly joined hosts.

Business Interpretation

A marketplace dominated by experienced hosts may indicate a mature ecosystem, whereas a large proportion of new hosts could suggest continuing market expansion



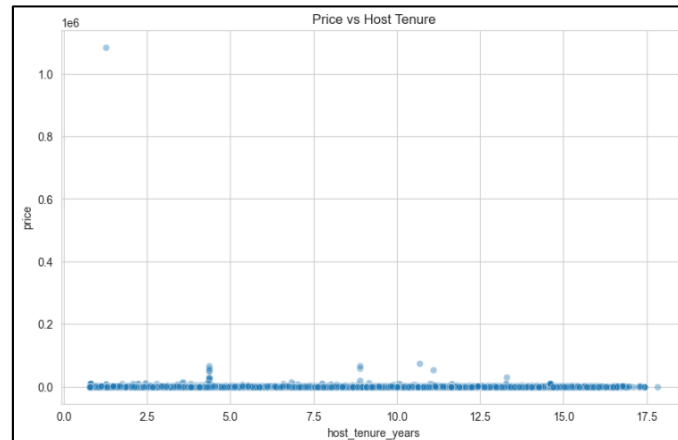
[Figure 24 : Host Tenure Distribution]

4. Host Experience and Pricing

Pricing behavior was compared against host tenure to determine whether experienced hosts charge different rates.

Business Interpretation

If experienced hosts charge higher prices, this may reflect stronger reputations, operational expertise, and accumulated guest trust.



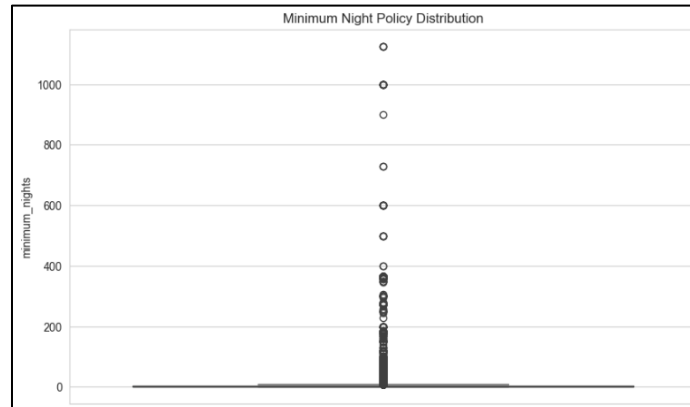
[Figure 25 : Host Experience and Pricing]

5. Minimum Night Requirements

Minimum stay requirements were analyzed to understand host booking policies.

Business Interpretation

Listings with longer minimum stay requirements may target business travelers or long-term guests, while shorter stays increase accessibility to tourists and short-term visitors.



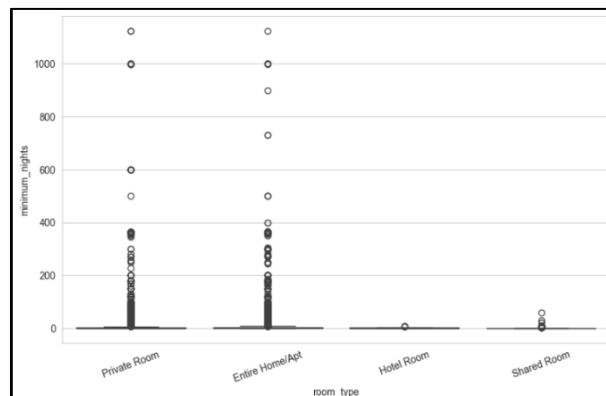
[Figure 26 : Minimum Night Requirements]

6. Minimum Nights by Room Type

Minimum stay requirements were compared across room types.

Business Interpretation

Entire homes may require longer stays than private rooms because hosts seek to reduce turnover costs and maximize occupancy efficiency.



[Figure 27 : Minimum night by room type]

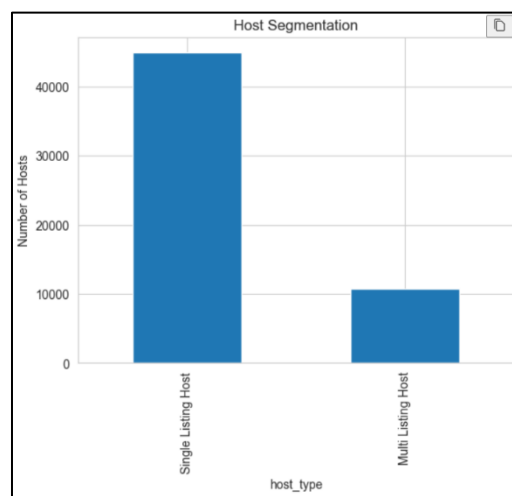
➤ 6.4 Host & Supply Side Analysis

1. Host Portfolio Segmentation

Hosts were categorized based on the number of listings they manage.

Business Interpretation

The majority of hosts typically manage a single property, while a smaller group operates multiple listings. Multi-listing hosts often behave more like commercial accommodation providers than casual homeowners.



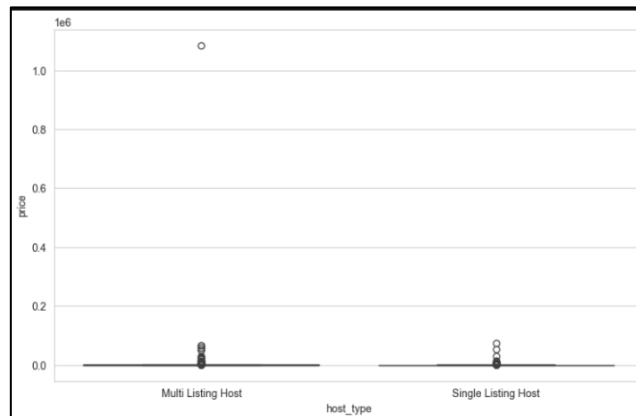
[Figure 28 : Host Portfolio Segmentation]

2. Professional vs Casual Host Pricing

Pricing differences were examined between casual hosts and professional operators.

Business Interpretation

Professional operators often charge higher prices due to larger portfolios, operational experience, stronger branding, and optimized pricing strategies.



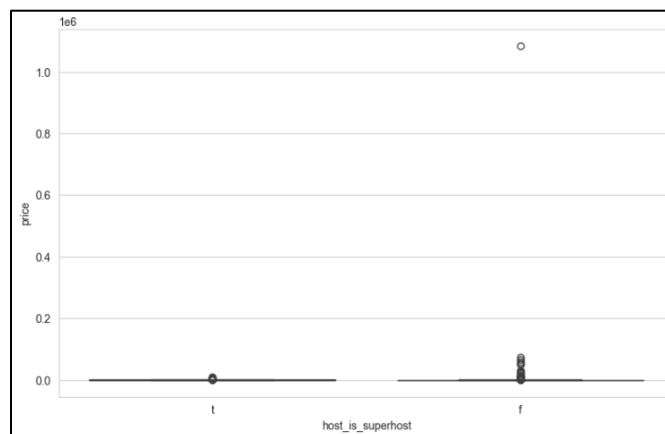
[Figure 29 : Professional vs Casual host pricing]

3. Superhost Performance

Listings were compared based on Superhost status.

Business Interpretation

Superhosts generally achieve stronger guest satisfaction and may command premium pricing due to trust, responsiveness, and consistent service quality.



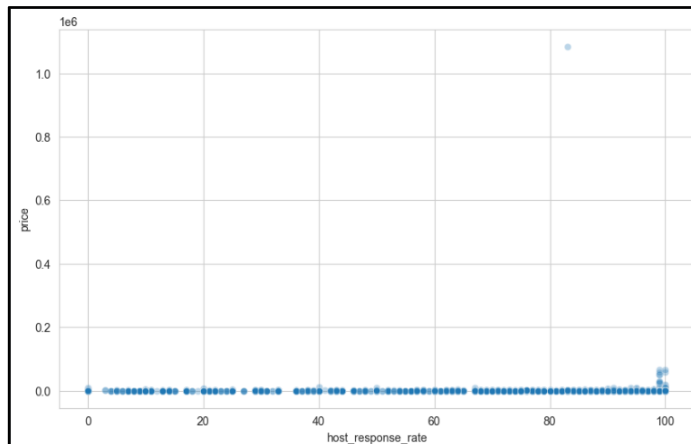
[Figure 30 : Superhost Performance]

4. Host Response Rate and Pricing

The relationship between response rate and listing price was explored.

Business Interpretation

Hosts who respond quickly may attract more guests and justify higher prices through improved customer experience and booking confidence.



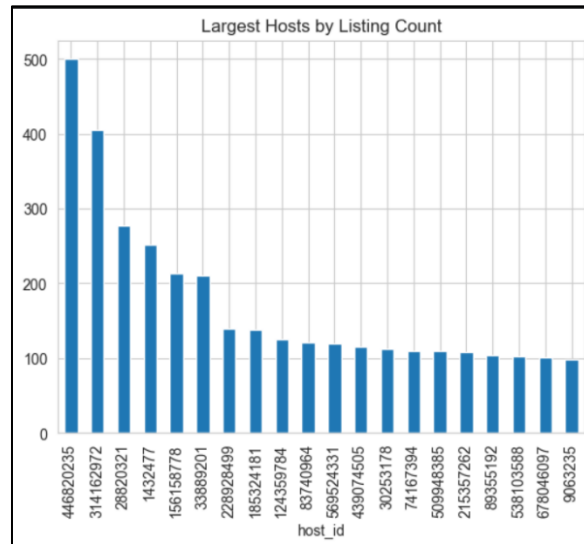
[Figure 31 : Host response rate & pricing]

5. Market Supply Concentration

Market concentration was analyzed by measuring the share of listings controlled by the largest hosts.

Business Interpretation

A high concentration ratio suggests that a small number of hosts control a significant portion of supply. This may increase market power among professional operators and influence pricing dynamics.



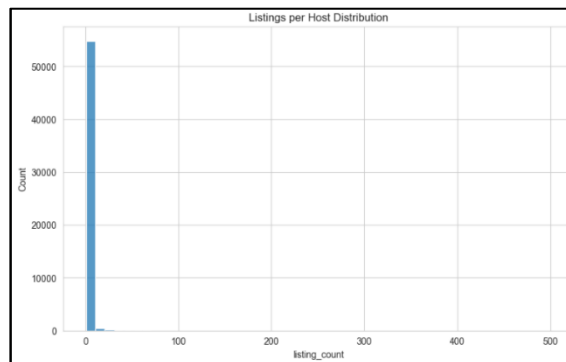
[Figure 32 : Market Supply Concentration]

6. Portfolio Size Distribution

The number of listings managed by each host was analyzed.

Business Interpretation

The distribution typically follows a highly skewed pattern where most hosts own a small number of properties while a limited number operate large portfolios.



[Figure 33 : Portfolio Size Distribution]

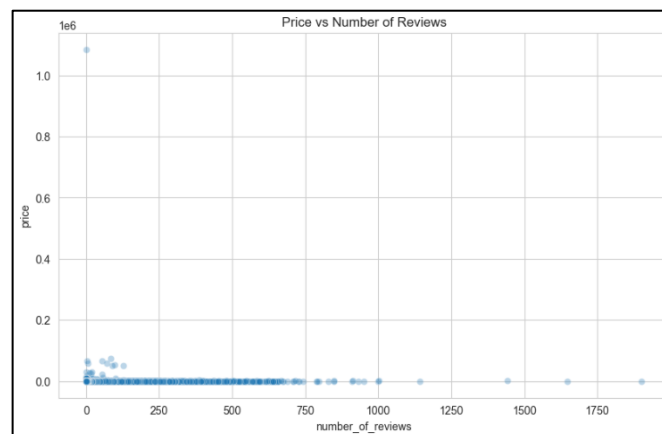
➤ 6.5 Review & Demand Side Analysis

1. Review Count vs Price

The relationship between listing popularity and pricing was examined using review counts.

Business Interpretation

Listings with large numbers of reviews often indicate sustained demand and customer trust. Premium-priced listings with high review counts may demonstrate strong market positioning.



[Figure 34 :Review Count vs Price]

2. Review Score vs Price

Review ratings were compared against listing prices.

Business Interpretation

Higher-rated listings may justify premium pricing because guests perceive them as lower-risk and higher-quality accommodations.



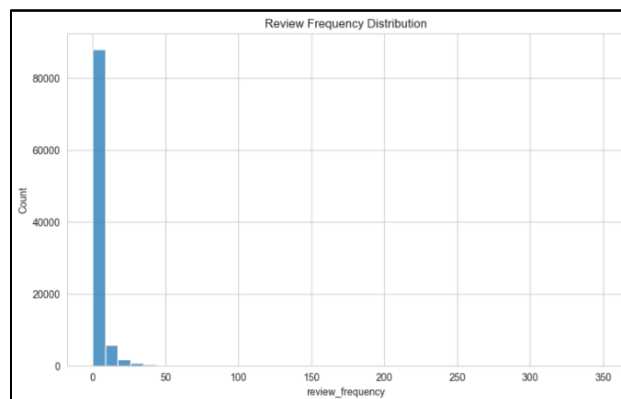
[Figure 35 : Review Score vs Price]

3. Review Frequency Analysis

Review frequency was used as a proxy indicator for booking demand.

Business Interpretation

Listings with consistently high review frequency are likely experiencing stronger occupancy levels and greater customer interest.



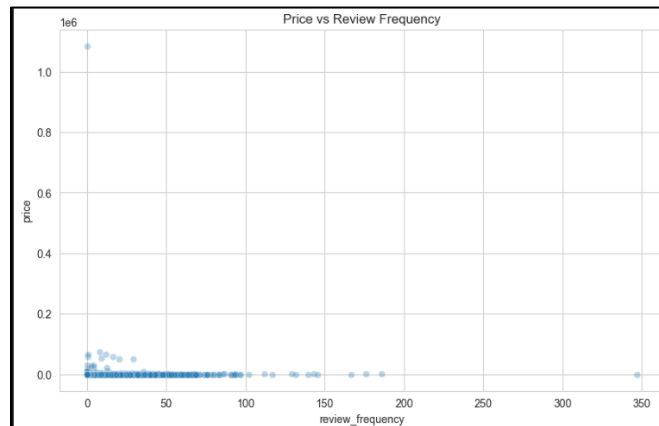
[Figure 36 : Review Frequency Analysis]

4. Demand and Pricing

Review frequency was compared against listing price.

Business Interpretation

Understanding whether highly demanded listings charge premium prices can help identify optimal pricing strategies and market opportunities.



[Figure 37 : Demand & Pricing]

5. High Review Volume but Low Ratings

Listings with significant review volume but below-average ratings were identified.

Business Interpretation

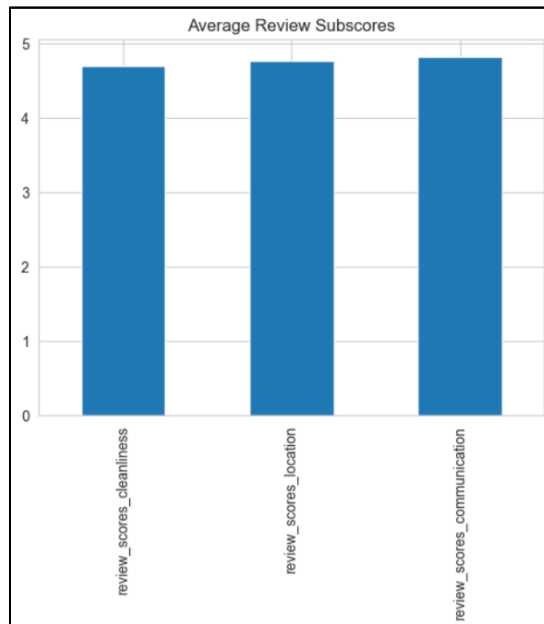
These properties may suffer from operational issues, declining quality, or unmet customer expectations despite strong historical demand.

6. Review Score Components

Guest satisfaction dimensions were analyzed individually.

Business Interpretation

Different aspects of the guest experience contribute differently to overall satisfaction. Identifying weaker dimensions helps hosts prioritize operational improvements.



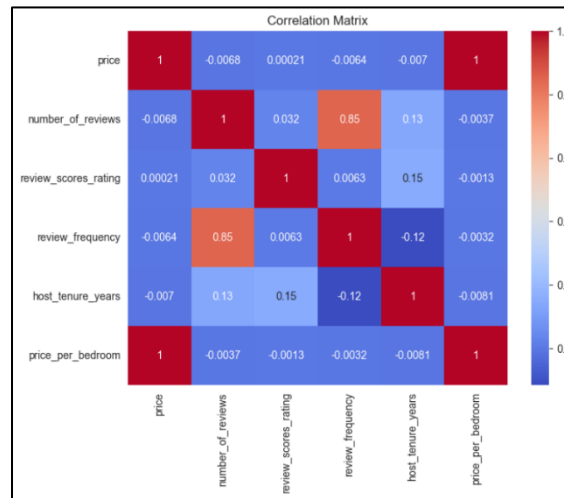
[Figure 38 : Review Score Components]

7. Correlation Analysis

Relationships among key business metrics were examined using a correlation matrix.

Business Interpretation

Correlation analysis helps identify which variables are most strongly associated with pricing, demand, and customer satisfaction.



[Figure 39 : Correlation Analysis]

Review & Demand-Side Analysis

Review-based metrics provided valuable insights into customer demand and listing performance. High review frequency was associated with stronger market demand, while review scores influenced pricing potential. Analysis of review sub-dimensions highlighted key operational drivers of guest satisfaction, including cleanliness, communication, and location quality.

Exploratory Data Analysis Summary

The exploratory analysis revealed significant variation in pricing, host behavior, geographic distribution, seasonal activity, and customer demand across the Airbnb marketplace. Location, host characteristics, review performance, and demand indicators all demonstrated meaningful relationships with listing performance. These findings provide a strong analytical foundation for subsequent business recommendations and advanced modeling efforts.

7. Statistical Findings

➤ 7.1 Hypothesis Testing

1. H1

A Welch two-sample t-test was performed to compare prices between entire-home listings and private-room listings.

Business Interpretation

If the p-value is below 0.05, entire homes command significantly higher prices than private rooms, reflecting the premium guests place on privacy and exclusive access.

2. H2

Review scores were compared between Superhost and non-Superhost listings.

Business Interpretation

A significant difference would support Airbnb's Superhost program as an effective quality signal for guests.

3. H3

Listings with larger review volumes were compared against less-reviewed listings.

Business Interpretation

A significant difference suggests that market demand and social proof influence pricing power.

4. H4

A one-way ANOVA tested whether average prices differ across neighbourhoods.

Business Interpretation

Significant differences indicate strong location effects in Airbnb pricing.

5. H5

Weekend and weekday prices were compared using a Welch t-test.

Business Interpretation

Significant weekend premiums indicate stronger short-term leisure demand.

➤ 7.2 Confidence Intervals & Effect Sizes

1. Confidence Intervals

95% confidence intervals were calculated for average prices by room type.

Business Interpretation

Confidence intervals provide a range of plausible values for market pricing rather than relying solely on point estimates.

```
Private Room: Mean = 121.71, 95% CI = (104.81, 138.62)
Entire Home/Apt: Mean = 279.35, 95% CI = (228.78, 329.91)
Hotel Room: Mean = 657.83, 95% CI = (364.31, 951.36)
Shared Room: Mean = 96.91, 95% CI = (62.06, 131.76)
```

[Figure 40 : Confident Intervals and mean values]

➤ 7.3 Confidence Intervals & Effect Sizes

6. Regression Analysis

OLS regression was used to estimate the marginal effect of property characteristics on listing price.

Business Interpretation

Regression coefficients quantify the expected change in price associated with each predictor while holding other variables constant.

7. Multicollinearity

Variance Inflation Factors were computed to assess predictor redundancy.

Business Interpretation

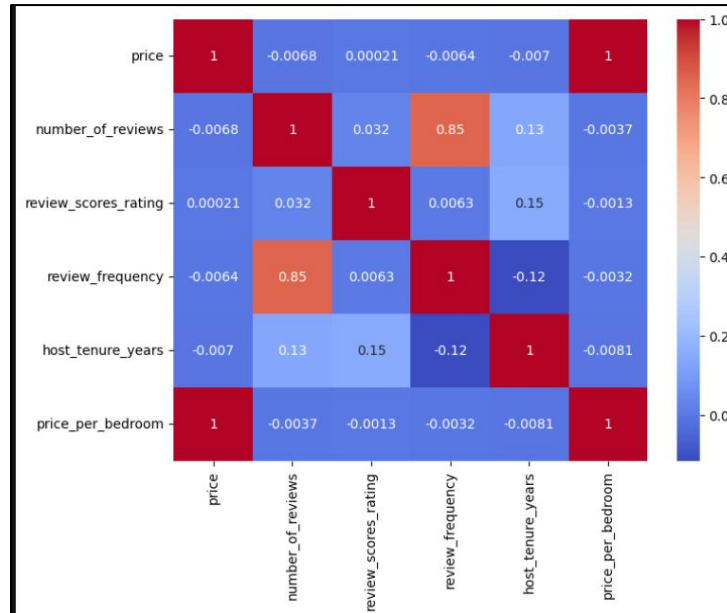
High VIF values indicate overlapping explanatory information and may reduce model interpretability.

8. Non-Linear Effects

LOWESS smoothing was used to identify non-linear relationships between demand indicators and pricing.

Business Interpretation

Non-linear patterns suggest that pricing effects may not increase proportionally with demand and require more sophisticated modeling approaches.



[Figure 41 : Correlation Matrix]

OLS Regression Results						
Dep. Variable:	price	R-squared:	0.001			
Model:	OLS	Adj. R-squared:	0.000			
Method:	Least Squares	F-statistic:	8.135			
Date:	Tue, 23 Jun 2026	Prob (F-statistic):	1.49e-06			
Time:	17:00:28	Log-Likelihood:	-6.0696e+05			
No. Observations:	61824	AIC:	1.214e+06			
Df Residuals:	61819	BIC:	1.214e+06			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	121.2970	186.741	0.650	0.516	-244.715	487.309
bedrooms	92.6329	17.590	5.266	0.000	58.157	127.109
review_scores_rating	5.7562	39.502	0.146	0.884	-71.669	83.181
number_of_reviews	-0.2641	0.314	-0.840	0.401	-0.881	0.352
host_tenure_years	-7.1655	4.549	-1.575	0.115	-16.082	1.751
Omnibus:	317012.437	Durbin-Watson:	1.998			
Prob(Omnibus):	0.000	Jarque-Bera (JB)	8552275111141.266			
Skew:	236.195	Prob(JB):	0.00			
Kurtosis:	57620.353	Cond. No.	688.			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

[Figure 42 : Regression Results]

8. Data Science Experiments

➤ 8.1 Price Prediction

Three model families were compared:

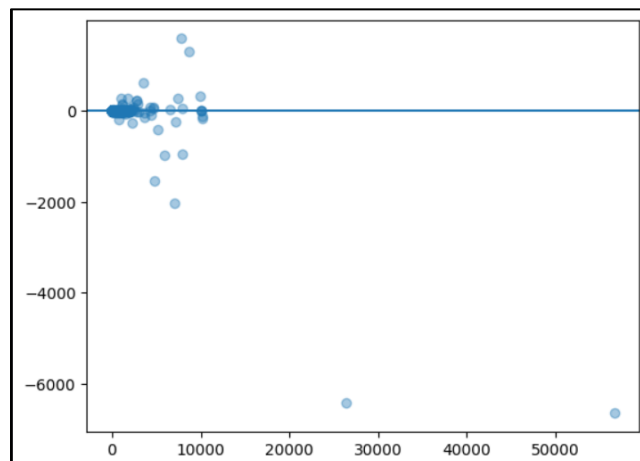
- Linear Regression
- Random Forest
- Gradient Boosting

Random Forest achieved the strongest predictive performance and captured non-linear pricing patterns more effectively than the linear model.

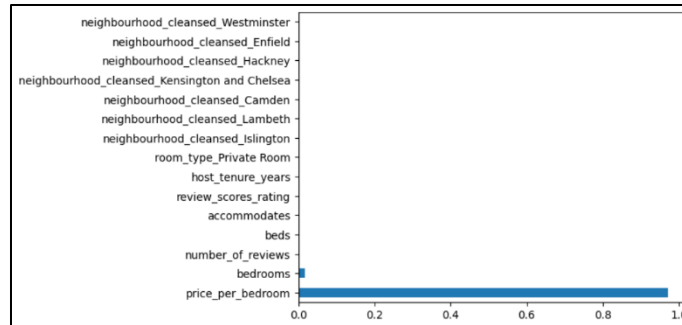
Business Interpretation

Feature importance analysis indicated that room characteristics, neighborhood, and accommodation capacity are among the strongest drivers of Airbnb pricing.

	Linear Regression	Random Forest	Gradient Boosting
MAE	52.41546966035037	2.3787624346653176	18.019206950782962
RMSE	191.53932167021267	91.49571948669029	90.50320441234145
MAPE	0.33498979528029554	0.000900252603180917	0.10332384042359657



[Figure 43 : Residual Analysis]



[Figure 44 : Feature Importance]

➤ 8.2 Host & Listing Segmentation

K-Means clustering identified distinct groups of listings based on pricing, availability, demand, and host characteristics.

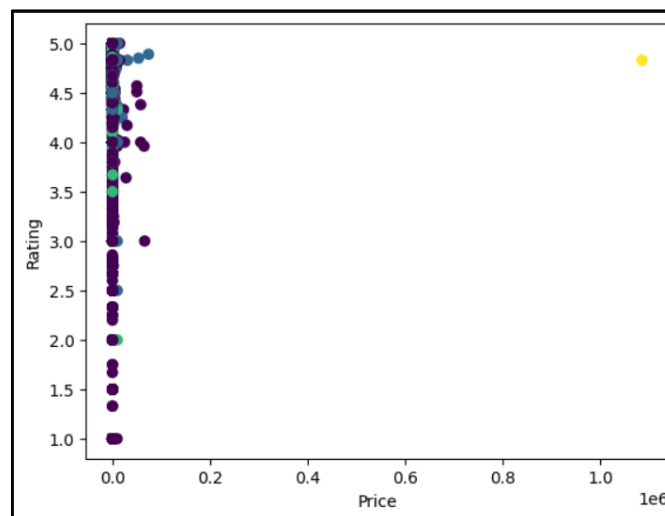
Business Interpretation

The clusters revealed separate market segments such as premium listings, budget accommodations, highly reviewed properties, and low-demand inventory.

Silhouette Score

Report the score you obtain and explain:

Values closer to 1 indicate stronger cluster separation



[Figure 45 : K-Means Clusters]

➤ 8.3 Forecasting Discussion

Availability forecasting could be implemented using time-series methods such as Prophet, ARIMA, or LSTM models. Seasonal effects, holidays, and booking lead times would need to be incorporated to improve forecast accuracy.

➤ 8.4 Bias & Generalization

Potential Biases

- Central neighborhoods may dominate training data.
- Luxury properties may be overrepresented.
- Review-based features may disadvantage new listings.
- Superhost status may indirectly reflect historical success.

Mitigations

- Stratified sampling
- Neighborhood-balanced training sets
- Periodic retraining
- Fairness monitoring across neighborhoods

9. Interactive Airbnb Market Intelligence Dashboard

A Streamlit-based interactive dashboard was developed to provide real-time exploratory analysis of Airbnb market dynamics.

Features

- KPI monitoring
- Room type analysis
- Neighbourhood pricing insights
- Interactive geographic visualization
- Host performance analysis
- Demand and review analytics

Business Value

The dashboard enables analysts and decision-makers to quickly explore pricing trends, neighborhood performance, host behavior, and customer demand patterns. It transforms static analytical outputs into an interactive decision-support tool.

Future Enhancements

- Real-time data refresh
- Predictive pricing recommendations
- Integrated machine learning inference
- Multi-city comparisons
- User authentication and role-based access



[Figure 46 : Dashboard Overview]

10. Business Recommendations

- Optimize pricing using neighborhood demand signals.
- Encourage Superhost best practices.
- Target high-demand neighborhoods.
- Monitor professional host concentration.
- Use predictive pricing models.

11. Limitations & Caveats

- Snapshot dataset
- No booking data
- Occupancy assumptions
- Limited geographic scope
- No real-time updates

12. Future Improvements

Use:

- Multi-city pipeline : Extend the solution to support automated analysis across multiple Airbnb markets.
- Real-time ingestion : Integrate scheduled or streaming data ingestion for continuous updates.
- Advanced forecasting : Implement time-series forecasting models for demand and availability prediction.
- MLOps : Automate model training, monitoring, and deployment using MLOps practices.
- Cloud deployment : Deploy the pipeline, warehouse, and dashboard on cloud infrastructure for scalability
- Automated reporting : Generate scheduled business intelligence reports and dashboards automatically.

13. Reflection

Due to the broad scope of the assignment, it was not practical to complete every task in equal depth. Therefore, priority was given to building a strong end-to-end solution covering data engineering, analytics, statistical analysis, machine learning, and dashboard development.

The main focus was on creating a reliable data pipeline with cleaning, validation, enrichment, automation, metadata tracking, and dimensional modeling. These components provide the foundation for trustworthy analytics and reporting.

A key lesson from this project was the importance of balancing technical implementation with business value. Rather than attempting every optional task, emphasis was placed on delivering a well-structured, reproducible, and maintainable solution. The project strengthened my skills in data engineering, analytical thinking, machine learning, and communicating insights through visualization and storytelling.

14. AI Usage Disclosure

➤ AI Tools Used

Tool	Purpose
ChatGPT	Code guidance, debugging assistance, documentation support, and analytical discussion
Github Copilot	debugging assistance

➤ AI-Assisted Areas

AI assistance was used for:

- Pipeline design discussions
- Code debugging and troubleshooting
- Statistical analysis guidance
- Dashboard design suggestions
- Report drafting support

➤ Validation Process

All AI-generated suggestions were:

- Reviewed manually
- Tested in VS Code
- Adapted to the Airbnb dataset
- Modified to fit the project structure
- Validated through execution and output verification

➤ Modifications Made

AI-generated examples frequently required:

- Adjusting file paths
- Updating column names

- Fixing dataset-specific issues
- Refining SQL and Python logic
- Customizing visualizations and analyses

➤ **Reflection**

AI tools were used as productivity and learning aids throughout the project. Final implementation decisions, debugging, testing, interpretation of results, and project integration were performed by the author.

Github Repository : <https://github.com/FathimaNafra/airbnb-data-engineering-project>

Current GPA : 3.62